

Smart Asset Management and Resource Allocation Platform

Team Members

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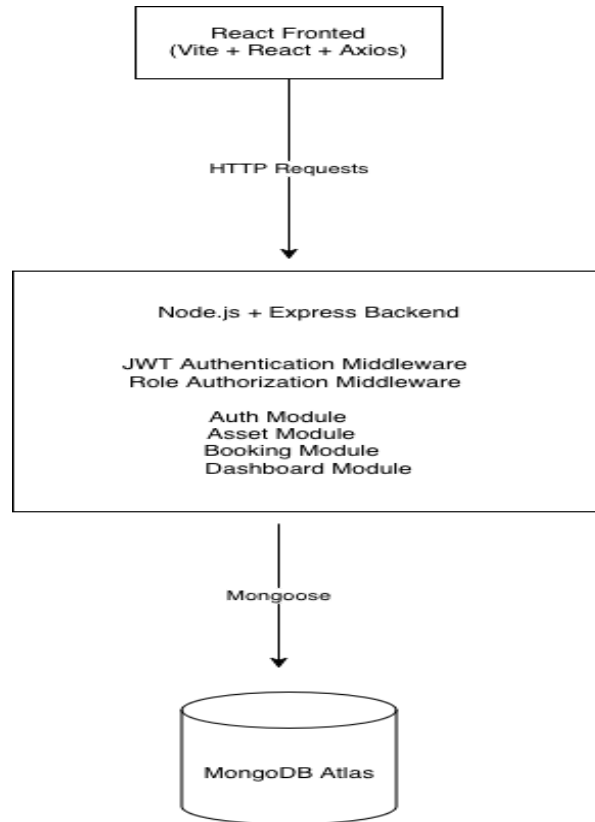
Problem Understanding

Organizations often manage shared resources using spreadsheets and manual processes, leading to booking conflicts, poor inventory visibility, delayed returns, and inefficient resource allocation. The proposed system provides a centralized platform for inventory management, asset booking, approval workflows, asset tracking, and analytics.

Objectives

- Inventory Management
- Asset Booking
- Approval Workflow
- Resource Allocation
- Utilization Tracking
- Analytics Dashboard

System Architecture



Architecture Description

Frontend:

- React.js
- Vite
- Axios

Backend:

- Node.js
- Express.js

Database:

- MongoDB Atlas

Authentication:

- JWT Authentication

Flow:

- User → React Frontend → Express APIs → MongoDB Database

Mention role-based access:

- Admin
- User

Database Schema Based on implemented models.

User Collection

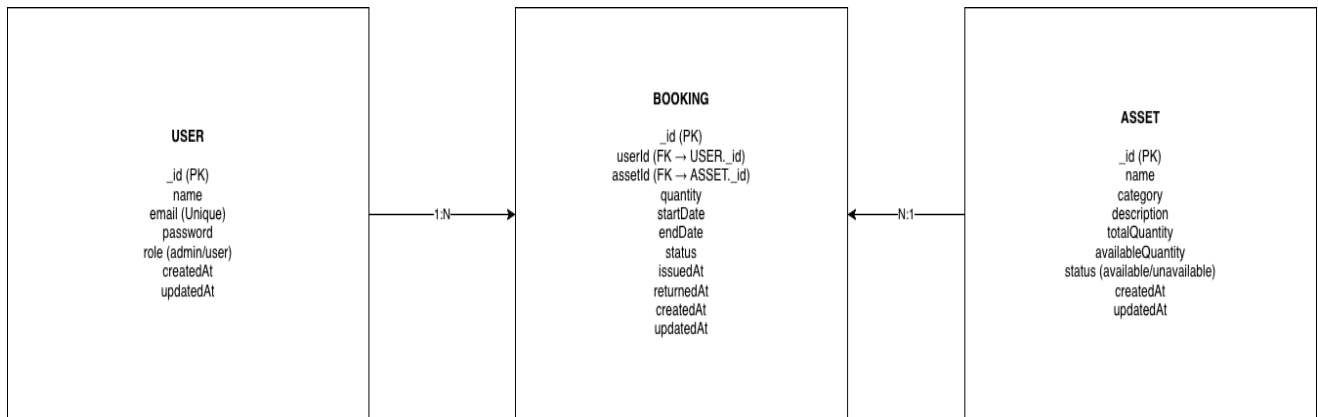
Field	Type
__id	ObjectId
name	String
email	String
password	String
role	String

Asset Collection

Field	Type
name	string
category	string
description	string
totalQuantity	Number
availableQuantity	Number
status	String

Field	Type
userId	ObjectId
assetId	ObjectId
quantity	Number
startDate	Date
endDate	Date
status	String
issuedAt	Date
returnedAt	Date

ER Diagram



Relationships

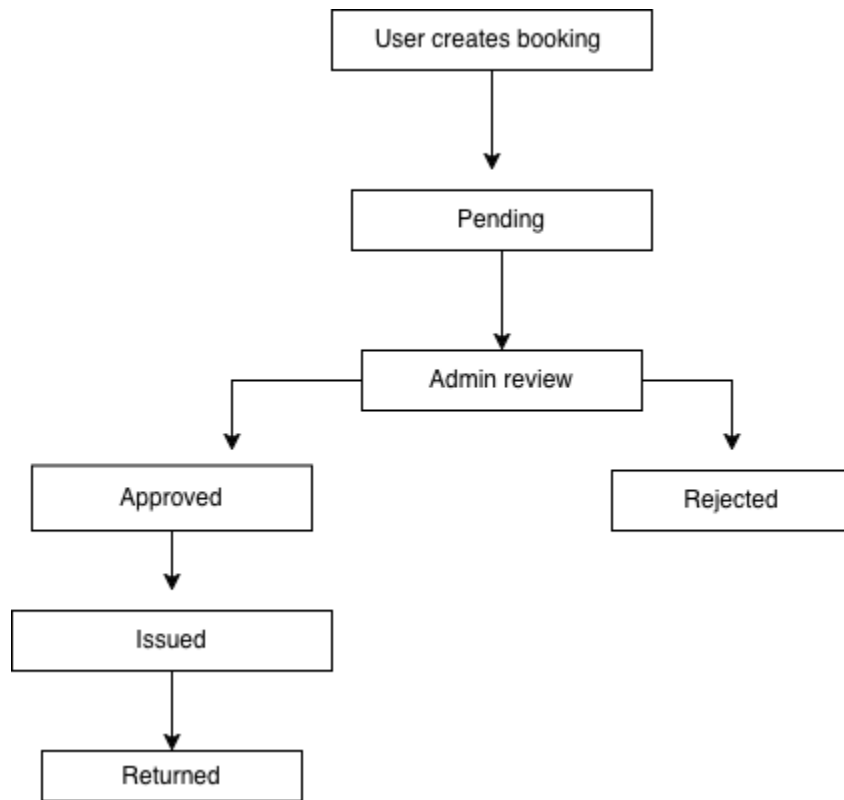
User (1) → Booking (Many)

Asset (1) → Booking (Many)

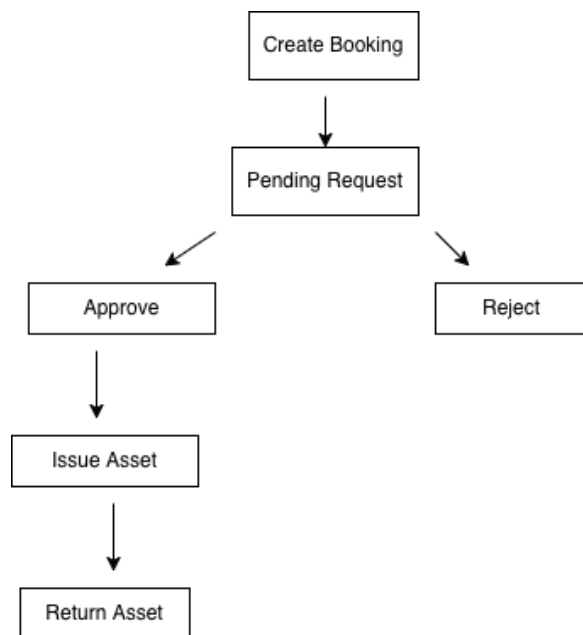
Relationship Summary:

User (1) ----- (Many) Booking (Many) ----- (1) Asset

Functional Workflow



API Overview



Method	Endpoint	Purpose
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POST	auth/register	Register User
POST	/auth/login	Login User
GET	/auth/profile	Get User Profile
GET	/assets	View Assets
POST	/assets/add	Add Asset
PUT	/assets/:id	Update Asset
DELETE	/assets/:id	Delete Asset
POST	/bookings	Create Booking
GET	/bookings/my-bookings	My Bookings
GET	/bookings/history	Borrowing History
GET	/bookings	All Bookings (Admin)
PUT	/bookings/:id/approve	Approve Booking
PUT	/bookings/:id/reject	Reject Booking
PUT	/bookings/:id/issue	Issue Asset
PUT	/bookings/:id/return	Return Asset
GET	/dashboard/summary	Dashboard Statistics
GET	/dashboard/most-utilized-assets	Asset Analytics
GET	/dashboard/overdue-assets	Overdue Assets

Features Implemented

Authentication

- User Registration
- Login
- JWT Security
- Role-Based Access Control

Inventory Management

- Add Asset
- Update Asset
- Delete Asset
- View Assets

Booking System

- Create Booking
- View Bookings
- Borrowing History

Approval Workflow

- Approve Request
- Reject Request

Asset Lifecycle

- Issue Asset
- Return Asset

Analytics Dashboard

- Dashboard Summary
- Most Utilized Assets
- Overdue Assets

Design Decisions & Conclusion

Design Decisions

MongoDB

Flexible schema

Easy scaling

Fast document storage

JWT

Stateless authentication

Secure API access

Separate Booking Collection

Maintains booking history

Supports workflow tracking

Simplifies analytics generation

Future Enhancements

- Email Notifications
- Audit Logs
- QR Code Asset Tracking
- Asset Health Monitoring
- Docker Deployment